CSCI 5526

PyTorch Tutorial

Announcements

- MP2 due Friday!
- Project groups must be finalized by 10/16!

Student Questions

I'm not sure but what do you think to what extent causal inference can ever fully replace randomized clinical trials?

Causal inference cannot replace RCTs ever. RCTs are still the gold-standard to establish the efficacy of any intervention.

However, causal inference excels in some aspects:

1. When the interventions are harmful or happen over a long time
2. Typically, low cost because we use pre-existing data
3. High chance of generalizability since there are no strict inclusion/exclusion criteria like in RCTs

Student Questions

To what extent can casual inference improve healthcare decision-making ?

It transforms real world data analysis from simple association studies

RCTs typically are very selective, we can apply causal inference to larger populations, especially specific excluded groups.

Also, evaluating comparative effectiveness is also costly in RCTs.

Student Questions

When using the parametric g-formula, how sensitive is the estimated ATE to the choice or misspecification of the outcome model f(x,t), how does poor covariate overlap between treated and control patients affect the estimate, how can we assess whether the counterfactual predictions are reliable, and what can we do if these assumptions appear to be violated?

1. It is highly sensitive to the choice of the model. For example, a high-bias model could lead to errors that do not vanish with more data.
2. Poor covariate overlap also is problematic as it violates common support assumption. The predictions can be artifactual and rely a lot on the functional form of the model.
3. To ensure counterfactual predictions are reliable, you would do what you do to evaluate a typical ML model fit. (look at training error, test error, etc).

Student Questions

Are covariate methods (like Ancova) preferred over other ML models and why?

-- Not necessarily.

-- The preference depends entirely on whether your priority is strict mathematical estimation or handling high-dimensional, complex data relationships.

We assume that treatments are assigned somewhat randomly. Do we also need to assume there is a balance of treatments, or just a probability greater than 0?

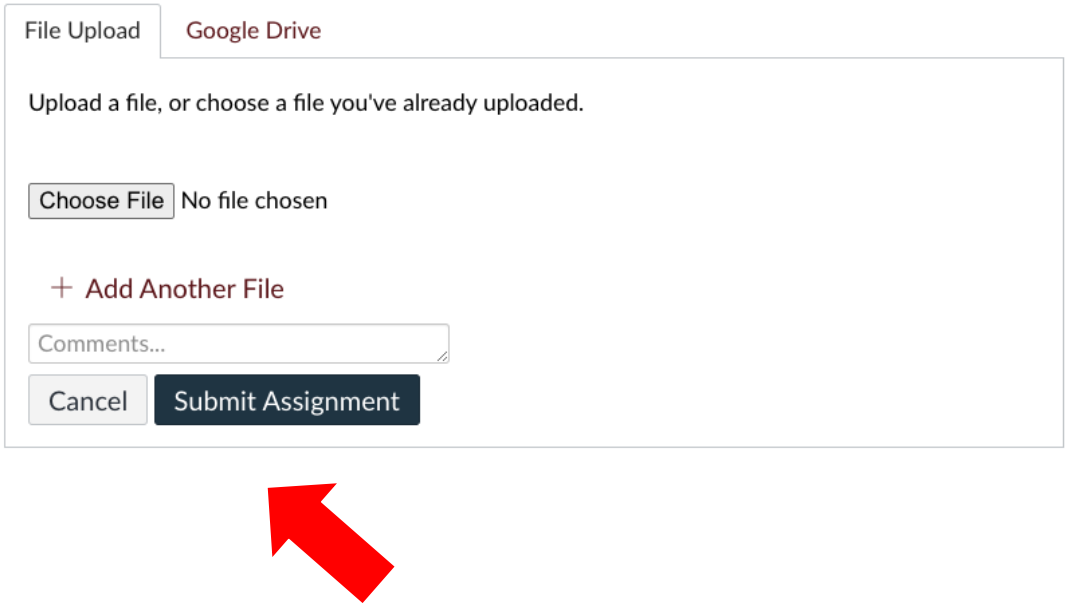
-- It is not required, but it is preferable to have a good balance between the treatment arms.

Outline

- PyTorch tutorial
  - Tensors
  - Tensor operations
  - Gradients
  - CPU vs GPU
  - A basic ML model
  - Training and evaluation

Submission

- Once finished, submit on Canvas (Lab 1).



Exit pass

https://z.umn.edu/ml4h_exitpass

